

Non-Uniform Fourier Infilling (NUFI): An Infinitely Differentiable, Covariance-Preserving Spectral Reconstruction Framework

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Abstract

Missing values in non-uniformly sampled, multi-dimensional time series complicate analysis because of irregular sampling and sensor dropout. Standard interpolation often introduces derivative discontinuities and distorts cross-channel covariance. We describe **Non-Uniform Fourier Infilling (NUFI)**, a GPU-accelerated framework that reconstructs missing values with a regularized trigonometric basis, yielding infinitely differentiable (C^∞) reconstructions. To preserve inter-channel correlations, we use a stochastic posterior injection method based on an LDL^T factorization of a stacked real-imaginary spectral covariance matrix. The framework uses a Type-1 Fast Non-Uniform Discrete Fourier Transform (NUDFT) with Gaussian gridding and an iterative Conjugate Gradient solver, giving $\mathcal{O}(N \log N)$ time complexity. The Tikhonov regularization parameter is selected automatically via Generalized Cross-Validation (GCV). Experimental validation is ongoing.

1 Introduction

Financial markets, robotics, and industrial IoT frequently produce multi-dimensional time-series data at irregular intervals. In such panels, missing data can degrade downstream models. Common imputation methods introduce several mathematical and statistical problems:

1. **Derivative Discontinuity:** Spline or linear interpolations introduce localized discontinuities in the first (C^1) or second (C^2) derivatives of the reconstructed signal. For physical systems where velocities and accelerations are important, such discontinuities can affect downstream prediction quality.
2. **Covariance Distortion:** Imputing columns independently can distort the cross-channel correlation structure, potentially damping joint variance and altering the eigenvectors of the empirical covariance matrix.
3. **Ill-Conditioning:** High-frequency localized sampling clusters can cause standard spectral estimation techniques [3, 4] to produce large oscillations, leading to numerical instability.

We introduce **NUFI**, a spectral infilling library built on PyTorch (with Metal Performance Shaders and CUDA acceleration) and multi-threaded Cython. NUFi aims to produce C^∞ -smooth reconstructions that preserve cross-signal correlation. The implementation employs a Type-1 Fast NUDFT with Gaussian gridding and analytic apodization deconvolution [1, 2]. This reduces the matrix-vector product complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log N)$. Because the reconstruction is a finite sum of smooth exponentials, it is $C^\infty(\mathbb{R})$ – a property that removes derivative discontinuities entirely.

2 Core Mathematical Formulation

Let a multi-dimensional time series be defined by a set of observations $y_n = x(t_n)$ sampled non-uniformly at timestamps $t_n \in \mathbb{R}$ for $n \in \{0, 1, \dots, N-1\}$. We model the underlying continuous-time signal $x(t)$ as a sum of M trigonometric basis functions:

$$x(t) = \text{Re} \left(\sum_{k=0}^{M-1} c_k e^{2\pi i f_k t} \right) \quad (1)$$

where $c_k \in \mathbb{C}$ are the spectral coefficients and $f_k \in \mathbb{R}$ the frequency coordinates.

2.1 The Non-Uniform Discrete Fourier Transform (NUDFT)

The standard type-1 non-uniform discrete Fourier transform gives a direct estimate of the spectral coefficients:

$$\hat{c}_k = \frac{1}{N} \sum_{n=0}^{N-1} y_n \exp(-2\pi i t_n f_k) \quad (2)$$

In practice, we bound frequencies by the inverse of the smallest sampling interval, a heuristic that approximates the Nyquist limit for the finest local resolution:

$$f_{\max} = \frac{1}{2 \min_n (t_{n+1} - t_n)} \quad (3)$$

3 Joint Multi-Channel Covariance Preservation

When infilling multi-dimensional channels, solving the spectral coefficients independently distorts the inter-channel correlation. To enforce consistency without corrupting the deterministic fit, we introduce a stochastic posterior injection that approximates the observed correlation structure.

Definition 1 (Spectral Covariance Matrix). *For each channel $d = 1, \dots, D$, let $\hat{\mathbf{c}}^{(d)} \in \mathbb{C}^M$ be the vector of regularized spectral coefficients. Define the stacked real-imaginary vector $\mathbf{z}_k = (\Re(c_k^{(1)}), \Im(c_k^{(1)}), \dots, \Re(c_k^{(D)}), \Im(c_k^{(D)}))^\top \in \mathbb{R}^{2D}$ for frequency index k . The joint (empirical) spectral covariance matrix $\Sigma \in \mathbb{R}^{2D \times 2D}$ is the sample covariance of $\{\mathbf{z}_k\}_{k=0}^{M-1}$.*

3.1 LDL^T Structural Factorization

To generate correlated noise, we compute the pivoted LDL^T decomposition of Σ :

$$\mathbf{P}\Sigma\mathbf{P}^T = \mathbf{L}\mathbf{D}\mathbf{L}^T \quad (4)$$

where $\mathbf{P}$ is a permutation matrix, $\mathbf{L} \in \mathbb{R}^{2D \times 2D}$ is unit lower triangular, and $\mathbf{D} = \text{diag}(d_1, \dots, d_{2D})$ contains the orthogonalized variances.

3.2 Stochastic Posterior Injection

Rather than perturbing the deterministic base signal $\tilde{\mathbf{x}}(t)$, we inject correlated noise onto the posterior. Let $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{2D})$ be a standard normal vector. The correlated noise vector $\mathbf{Z} \in \mathbb{R}^{2D}$ is obtained by

$$\mathbf{Z} = (\mathbf{P}^T \mathbf{L} \mathbf{D}^{1/2}) \boldsymbol{\eta} \quad (5)$$

so that $\text{Cov}(\mathbf{Z}) = \Sigma$. We then scale each component i by the ratio of the observed residual standard deviation $\sigma_{\text{resid},i}$ to the square root of the corresponding diagonal entry of Σ :

$$Z_{\text{scaled},i} = \frac{\sigma_{\text{resid},i}}{\sqrt{\Sigma_{ii}}} Z_i \quad (i = 1, \dots, 2D). \quad (6)$$

The final infilled values are the sum of the deterministic reconstruction and this scaled correlated noise:

$$\mathbf{x}_{\text{infilled}}(t) = \tilde{\mathbf{x}}(t) + \mathbf{Z}_{\text{scaled}}(t) \quad (7)$$

By construction, this procedure approximately preserves the channel-wise correlation structure while matching the marginal residual volatilities.

4 Advanced Optimization and Solvers

4.1 Tikhonov Regularized Formulation

To reduce ill-conditioning and numerical instability from clustered timestamps, we formulate the infilling problem as a regularized least-squares optimization over the complex coefficients $\mathbf{c}$:

$$\min_{\mathbf{c}} \|\mathbf{A}\mathbf{c} - \mathbf{y}\|_2^2 + \alpha \|\mathbf{c}\|_2^2 \quad (8)$$

where $\mathbf{A} \in \mathbb{C}^{N \times M}$ is the forward mapping matrix with elements $A_{n,k} = \exp(2\pi i f_k t_n)$, and $\alpha > 0$ is the Tikhonov (L_2) regularization parameter [6].

4.2 SVD-Based Generalized Cross-Validation (GCV)

The selection of α and M is automated using Generalized Cross-Validation (GCV) [8]. The GCV score can be evaluated efficiently after an SVD of $\mathbf{A}$.

Proposition 1 (GCV score via SVD). *Given the SVD $\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^H$ with singular values σ_j , let $\tilde{\mathbf{y}} = \mathbf{U}^H \mathbf{y}$. The GCV score $V(\alpha)$ can be evaluated in $\mathcal{O}(M)$ time as:*

$$V(\alpha) = \frac{\frac{1}{N} \left[\sum_{j=1}^M \left(\frac{\alpha}{\sigma_j^2 + \alpha} \right)^2 |\tilde{y}_j|^2 + \|\mathbf{y}\|_2^2 - \|\tilde{\mathbf{y}}\|_2^2 \right]}{\left(1 - \frac{1}{N} \sum_{j=1}^M \frac{\sigma_j^2}{\sigma_j^2 + \alpha} \right)^2} \quad (9)$$

Proof. Let $\mathbf{H}(\alpha) = \mathbf{A}(\mathbf{A}^H \mathbf{A} + \alpha \mathbf{I})^{-1} \mathbf{A}^H$ be the influence matrix. The GCV score is

$$V(\alpha) = \frac{\frac{1}{N} \|(\mathbf{I} - \mathbf{H}(\alpha))\mathbf{y}\|_2^2}{\left(\frac{1}{N} \text{Tr}(\mathbf{I} - \mathbf{H}(\alpha)) \right)^2} \quad (10)$$

Expressing the numerator via SVD:

$$\|(\mathbf{I} - \mathbf{H}(\alpha))\mathbf{y}\|_2^2 = \sum_{j=1}^M \left(1 - \frac{\sigma_j^2}{\sigma_j^2 + \alpha} \right)^2 |\tilde{y}_j|^2 + \|\mathbf{y}\|_2^2 - \|\tilde{\mathbf{y}}\|_2^2 \quad (11)$$

For the denominator, the trace is:

$$\text{Tr}(\mathbf{I} - \mathbf{H}(\alpha)) = N - \sum_{j=1}^M \frac{\sigma_j^2}{\sigma_j^2 + \alpha} \quad (12)$$

Substituting the trace and numerator back yields the result. This formulation avoids repeated matrix inversions, making the hyperparameter search $\mathcal{O}(M)$ after a single SVD computation on a subsampled representation. $\square$

4.3 Scalable Conjugate Gradient (CG) Solver

For high-dimensional systems ($N > 10^4$), dense SVD becomes prohibitive ($\mathcal{O}(N^3)$). We therefore implement an iterative complex Conjugate Gradient (CG) solver. Each iteration requires two matrix-vector products with $\mathbf{A}$ and $\mathbf{A}^H$ ($\mathcal{O}(NM)$); convergence often requires $K \ll M$ iterations, and the solver is well-suited for GPU acceleration.

5 System Architecture and Automation

To support integration into automated data pipelines, NUFI offers the following features:

- **Automated Entrypoint:** The `impute_dataframe` function infers timestamps, handles column types, masks missing values, and selects hyperparameters via SVD-based GCV without manual intervention.
 - **JSON-Serializable Diagnostics:** The API returns detailed logs, including Signal-to-Noise Ratio (SNR), spectral entropy, and numerical stability flags.
 - **Snapshot-Based Version Control:** To preserve data lineage, the `TransformationTracker` captures compressed `.parquet` snapshots of dataframes before and after transformation in `.nufi_history/`. Reversion and recovery are supported by a Git/DVC-style interface.
 - **Transformation Logging and Safety Limits:** The tracking system uses append-only transformation logging. If a write operation fails, the tracker raises a `TransformationLoggingError`, halting the transaction to prevent silent data corruption.
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6 Experimental Evaluation

We compare NUFI to standard time-series imputation baselines: Cubic Spline interpolation, Multivariate Imputation by Chained Equations (MICE) [7], and Gaussian Process (GP) regression with a Radial Basis Function (RBF) kernel.

6.1 Setup and Metrics

The methods are evaluated on a synthetic multi-channel time-series dataset with $N = 250$ non-uniform timestamps, $D = 4$ correlated channels, and a missingness rate of 30%. Performance is measured via:

1. **Root Mean Squared Error (RMSE):** Reconstruction accuracy relative to the unmasked ground truth.
2. **Covariance Discrepancy (Frobenius Norm):** $E_{\text{cov}} = \|\Sigma_{\text{true}} - \Sigma_{\text{infilled}}\|_F$
3. **Runtime:** Wall-clock time on a single CPU core.

6.2 Current Status

Full experimental results are under preparation. A reproducible evaluation suite will be released together with the software.

7 Conclusion

We described NUFI, a library for infilling non-uniformly sampled multi-dimensional time series. By combining a type-1 Fast NUDFT with Tikhonov regularization, stochastic post-hoc covariance injection, automated GCV, and an iterative CG solver, NUFI delivers reconstructions with C^∞ smoothness and aims to preserve the empirical covariance. Future work will address experimental validation, non-stationary frequency estimation, and cooperative multi-agent parallel workflows.

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